



tomation of production and research and development.

"Another important trend is the modular architecture of the automation software," says Jain. "This brings efficiency by allowing you to reuse pre-written code or write new software modules so that these can be re-used," he adds.

Easing it up with automated configuration

Since most of the equipment is moving towards mobile control, calling for proper interfaces is important. Proper connectivity between test equipment and the computing device is very important. Several equipment call for installing specific drivers to get the system ready for work. This might go wrong in any number of steps.

BenchVue by Keysight has been a welcome change in this regard. By plugging an instrument into their PC over LAN, GPIB or USB, the instrument is automatically configured for use in BenchVue.

NI's TestStand also allows for a similar feature to reduce hassle. Some other features include parallelisation and integration of codes written in multiple languages. This should be a significant advantage for people looking to build a team versed in different languages.

Working with multiple scripting languages

Jain says, "Support for multiple scripting languages is essential for acceptability of the test automation software." According to NI Automated Test Outlook 2016, a need for a single scripting language was pointed out.

Shirin Bardia, media relations specialist, National Instruments India, suggests, "A newer approach to test

Customised solutions

"Flexibility is low with off-the-shelf equipment," says Rohit Joshi, co-founder, Logic Fruit Technologies. For example, "With generic interfaces, the number of probes are limited." Electronics hardware systems should be adaptable based on usage. This calls for the test setup to be similarly adaptable. Off-the-shelf software has a limit with regards to the number of probes allowed. "However, field programmable gate array custom solutions can be programmed for more," he adds.

Joshi explains, "Field programmable gate arrays can be modified from the application program interface via VHIC hardware description language or Verilog programming." VHIC stands for very-high-speed integrated circuit.

system development that builds a heterogeneous system out of multiple languages, allowing a use of multiple systems that utilise the best of all languages to build powerful systems."

An example would be to use Python for scripting validation, while C# could be used to develop an object-oriented interface for hardware. "Since all languages are designed to tackle specific applications, using each to its strengths would ultimately save time and money," Bardia adds. However, this comes with its own set of problems, as user understanding between different languages might become an issue at a later stage.

"Test departments are now turning to commercial off-the-shelf test management software to act as a Rosetta Stone of sorts between different languages," says Bardia. This way, each of you can turn to what you know best without having to worry about how to communicate with other portions of code.

She adds, "Test managers can take advantage of the full power of a heterogeneous design." All the different modules can be created in different languages, making maintenance and upgrades much easier. R&S Quickstep at present supports C#, .net, C++ and Python scripts in an attempt to make the tool usable across multiple languages.

However, it brings along its own

set of complications. Achutanna says, "Adding heterogeneous language support in a single software increases complexity, software dependencies and reduces portability."

Add-on software

INERTIA is a test automation software designed as a software add-on for NI VeriStand to provide a seamless, integrated control solution for test-cell applications. INERTIA control add-on to NI VeriStand provides multi-mode PID support in developing a real-time software application for Ford. It allows control of a single loop to switch between temperature, pressure or flow mode dynamically as the system calls for it.

With similar solutions, we can look forward to improvements in test automation. Jain adds, "Test automation software uses add-on software in an encapsulated manner, and depending on the test plan, support for multithreading is an advantage when it comes to speed and efficiency."

The automated future

With the introduction of automation in testing, the past 20 or so years have seen a lot of advances in aiding humans. Additions in automation might have begun with adding connectivity among devices; however, increasing dependence on computers requires many new practices in place.

The features of scripting through multiple languages in order to bring out the best person for the job are just the beginning. At this rate, we could very well look to a future where some glitch does not end in your phone catching fire. **EFY**

Leela Krishna Achutanna, senior manager market development, aerospace and development, Rohde and Schwarz, highlights some of the improvements brought upon by recent development trends. Major ones are:

- Performance-optimised design for low overhead test execution
- Intrinsic parallelism for efficient use of test system resources
- Graphical editor for simple development of new test procedures
- Efficient and simple extension of available test functions